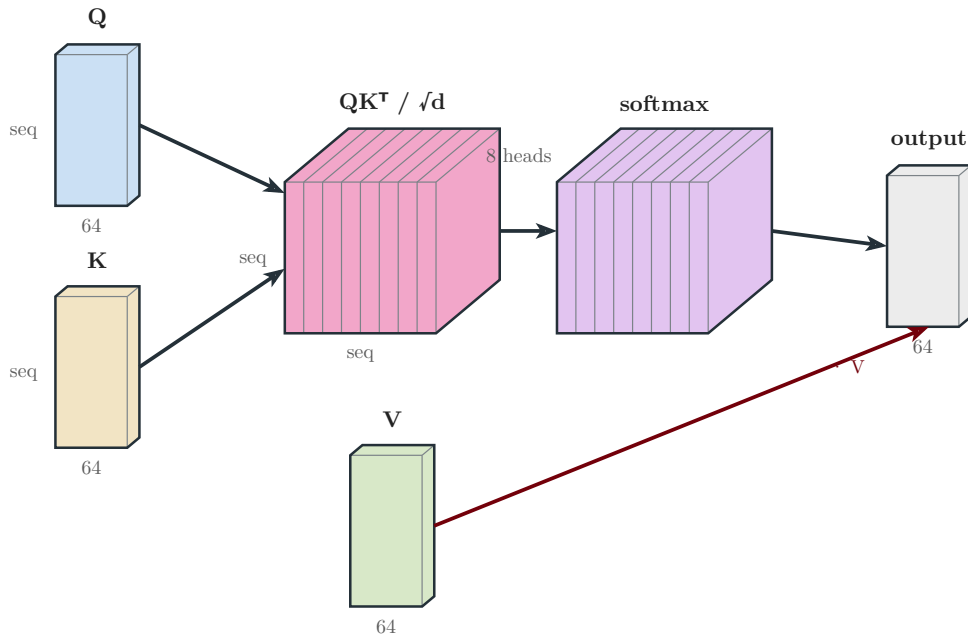


Multi-head self-attention block

$H = 8$ heads $\cdot d_k = d_v = 64$ $\cdot$ scaled dot-product attention $\text{softmax}(QK^\top / \sqrt{d_k})V$ per head



Q , K , V are linear projections of the input sequence ($\text{seq} \times d_k$). The score tensor $QK^\top / \sqrt{d_k}$ is a $\text{seq} \times \text{seq} \times 8$ cube — one $\text{seq} \times \text{seq}$ attention map per head (the depth seams). After softmax, each head's weights are applied to V ; the 8 head outputs are concatenated and linearly transformed to the block output.